

## ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

## Chapter 12: Inertial Reference Systems (IRS/INS) & Schuler Tuning

Syllabus Source: CAE Oxford (pp. 453-498) | Pass Target:  $\geq 90\%$  AviationExam

### 1. STRAPDOWN INERTIAL REFERENCE SYSTEMS (IRS)

Modern airliners use 'strapdown' IRS units with 3 orthogonal Ring Laser Gyroscopes (RLGs) and 3 quartz accelerometers fixed directly to the aircraft airframe:

- Ring Laser Gyro (RLG): Measures angular rate of rotation via the Sagnac Effect (interference between two counter-rotating laser beams). No moving mechanical parts, highly reliable.
- Accelerometer Integration: Accelerometers measure raw linear acceleration ( $a$ ). 1st integration yields VELOCITY ( $v = \int a \, dt$ ); 2nd integration yields POSITION ( $s = \int v \, dt$ ).
- Gyrocompassing Alignment: On the ground, while stationary for 5 to 10 minutes, the IRS senses the Earth's rotation ( $15^\circ/\text{hour}$ ) to automatically find TRUE NORTH and calculate local LATITUDE without external aid! Longitude MUST be manually entered by the pilot.

### 2. THE SCHULER TUNING PRINCIPLE (84.4 MINUTE PERIOD)

Because the Earth is curved, accelerating over a spherical surface would cause accelerometers to tilt and mistake gravity for acceleration, causing velocity errors to increase exponentially towards infinity:

- Schuler Pendulum: The IRS platform/computer is tuned to oscillate like a pendulum whose length equals the radius of the Earth ( $R = 6,371 \text{ km}$ ). Its natural oscillation period is:  $T = 2\pi \times \sqrt{R/g} = \text{EXACTLY } 84.4 \text{ MINUTES!}$
- Error Bounding: Schuler tuning forces velocity errors to oscillate harmlessly between positive and negative values every 84.4 minutes, preventing unbounded divergence!

### 3. IRS DRIFT & TRIPLE-MIX FMC INTEGRATION

- Bounded vs Unbounded Errors: Velocity and tilt errors oscillate with the 84.4 min Schuler cycle (bounded). Position errors creep forward with a steady DRIFT (unbounded, typically 1 to 2 NM per hour of flight).
- Triple Mix & Sensor Fusion: The Flight Management Computer (FMC) averages positions from 3 independent IRS units (Triple-Mix) and continuously updates the hybrid navigation solution using GPS and DME/DME updating.

### STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

#### [+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): IRS Alignment & Schuler Tuning

IRS Alignment Phases (Stationary on Ground):

1. Leveling: Accelerometers sense gravity vector  $g$  to find local vertical.
2. Gyrocompassing: Gyros sense horizontal component of Earth rotation ( $15^\circ/\text{h} \times \cos \text{lat}$ ) to find True North. Fails above  $78^\circ$ - $82^\circ$  latitude!
3. Position entry: Crew inserts precise gate coordinates.

Schuler Tuning:

- Oscillation period  $T = 84.4 \text{ minutes}$  (matches earth-radius pendulum).
- Prevents aircraft linear accelerations from corrupting vertical reference.
- Maximum allowable IRS drift: 1 to 2 NM per hour.

## 4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

### Q1: What is the natural period of the SCHULER OSCILLATION in an Inertial Reference System (IRS)?

- [A]: 24 hours
- [B]: EXACTLY 84.4 MINUTES
- [C]: 60 minutes
- [D]: 4 minutes

#### [?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Schuler period corresponds to an Earth-radius pendulum:  $T = 2\pi\sqrt{R/g} = 84.4$  minutes, bounding acceleration errors.

### Q2: During stationary ground alignment of an IRS, what parameters can the system determine completely auto

- [A]: Only magnetic variation
- [B]: TRUE NORTH and local LATITUDE (by sensing Earth's rotation rate and gravity)
- [C]: Longitude and altitude
- [D]: Runway heading

#### [?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Earth rotation vector (15°/h) provides True North and latitude =  $\arcsin(\text{vertical component} / 15^\circ)$ . Longitude has zero rotational signature and must be entered manually.

### Q3: What optical physical effect is utilized by Ring Laser Gyros (RLG) to measure aircraft angular rotation?

- [A]: Photoelectric effect
- [B]: THE SAGNAC EFFECT (frequency difference between counter-rotating laser beams)
- [C]: Hall effect
- [D]: Coriolis effect

#### [?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Rotation of the RLG alters path lengths of clockwise and counter-clockwise laser beams, generating an optical beat frequency proportional to turn rate.